



**COMSATS University Islamabad,
Abbottabad Campus**

SOFTWARE DESIGN DESCRIPTION

(SDD DOCUMENT)

for

KOJLO
Version 1.0

By

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Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) *include the ones given at scope time both in doc and presentation	Action Taken

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Signature_____

1.Introduction

Khojlo is a mobile and web-based business discovery platform designed to improve the visibility of newly established, affordable, and value-driven local businesses. The platform enables users to discover, search, compare, and explore businesses while allowing business owners to manage verified business profiles. Its primary objective is to bridge the gap between customers and local businesses by enhancing accessibility, increasing business visibility, and supporting informed decision-making.

The system provides user authentication and profile management, business registration and profile management, and a business discovery feed that highlights newly opened and trending businesses. It includes search, filtering, and comparison features that allow users to find businesses based on keywords, category, location, and price, and compare them based on pricing, services, and overall value.

Khojlo also incorporates a reviews and ratings system, Google Maps integration for location-based discovery and navigation, an AI chatbot that answers business-related queries using a Retrieval-Augmented Generation (RAG) approach, and an AI preference system that personalizes business recommendations based on user interests. In addition, the platform supports direct messaging between users and businesses and provides an administrative module for business verification, content moderation, and spam management.

The scope of the project is limited to business discovery, comparison, reviews and ratings, location-based services, personalized recommendations, AI-assisted business support, business information accessibility, and communication between users and businesses. Online ordering, payment processing, and delivery services are outside the scope of the system.

2.Design methodology and software process model

The Khojlo system follows the Object-Oriented Design (OOD) methodology. This methodology was selected because the system consists of multiple real-world entities such as users, business owners, businesses, reviews, ratings, offers, and administrators, which can be naturally represented as objects with their own attributes and behaviors.

Object-Oriented Design provides a modular structure where each component of the system can be developed, tested, and maintained independently. This improves code reusability, scalability, and maintainability while making future enhancements easier. Since the project is developed using Flutter for the frontend and FastAPI (Python) for the backend, both of which support object-oriented programming principles, the selected methodology is well suited to the overall system architecture.

The object-oriented approach also supports clear separation of responsibilities among system modules, making the overall design easier to understand and extend.

2.1 Software Process Model

The development of Khojlo follows the Agile Software Development methodology using the Scrum framework. Agile was selected because the project comprises multiple independent modules that can be developed and integrated incrementally. The development process is organized into iterative sprints, enabling each module to be designed, implemented, tested, and refined throughout the project lifecycle.

The Scrum framework supports incremental development, continuous testing and integration of new features, regular progress reviews and team meetings, efficient task distribution among team members, and the flexibility to accommodate changing requirements during development. This iterative approach allows the development team to monitor progress effectively, identify and resolve issues early, and deliver functional components at the end of each sprint rather than waiting until the completion of the entire project.

3. System overview

Khojlo is a cross-platform business discovery application that enables users to search, explore, compare, and interact with local businesses through both mobile and web platforms. The system primarily focuses on increasing the visibility of newly established and value-driven businesses while providing users with reliable business information in a structured and user-friendly environment.

The platform provides secure authentication for different user roles, including customers, business owners, and administrators. Business owners can register and manage verified business profiles, while users can discover businesses through search, filtering, comparison, reviews, ratings, interactive maps, and personalized discovery feeds. An AI-powered chatbot assists users by answering business-related questions using information stored within the platform. Additionally, the integrated messaging system allows users to communicate directly with business owners for inquiries, improving customer engagement and business accessibility.

The system adopts a client-server architecture in which the Flutter frontend communicates with a FastAPI backend through RESTful APIs. Business information, user accounts, reviews, messages, and other application data are securely stored in a PostgreSQL database. Google Maps services provide location-based search and navigation, while administrative tools ensure the authenticity of business listings and maintain overall platform quality through verification and content moderation.

The overall design emphasizes modularity, scalability, maintainability, and ease of future expansion while remaining focused on the primary objective of improving local business discovery and strengthening the connection between customers and business owners.

3.1 Architectural design

The Khojlo system follows a three-tier client-server architecture, which separates the application into the Presentation Layer, Application Layer, and Data Layer. This architecture promotes modularity, maintainability, scalability, and easier integration of external services. Each layer is responsible for a specific set of tasks and communicates with adjacent layers through well-defined interfaces.

The Presentation Layer consists of the Flutter-based mobile and web applications through which users, business owners, and administrators interact with the system. This layer is responsible for displaying information, collecting user input, and communicating with the backend using RESTful APIs.

The Application Layer is implemented using FastAPI and contains the business logic of the system. It processes client requests, performs authentication and authorization, manages business operations, handles reviews, favorites, messaging, chatbot requests, and administrative functions. This layer also communicates with external services such as Google Maps APIs and the AI chatbot module whenever required.

The Data Layer uses PostgreSQL to store and manage all application data, including user accounts, business profiles, reviews, ratings, favorites, promotions, chat messages, and administrative records. The backend retrieves and updates information in the database whenever users interact with the system.

External services such as Google Maps APIs provide mapping, navigation, and location-based functionality, while the RAG-based AI Chatbot retrieves business information stored within the platform database to answer user queries.

This modular decomposition ensures that changes in one layer have minimal impact on other layers, making the system easier to develop, maintain, and extend in the future.

3.1.1 Module Relationships

The system is divided into independent functional modules that work together to provide the overall functionality of Khojlo. The User Authentication Module validates users before granting access to protected features, while the Business Management Module enables business owners to register and manage business profiles. The Business Discovery Module retrieves newly added and trending businesses, and the Search, Filter, and Comparison Module processes user queries to return relevant business results. The Reviews and Ratings Module stores and displays customer feedback, whereas the Maps Integration Module retrieves business locations using Google Maps services.

The AI Chatbot Module processes business-related queries and retrieves relevant information from the platform database using a Retrieval-Augmented Generation (RAG) approach. The Chat and

Messaging Module facilitates direct communication between users and business owners, while the Admin Module is responsible for verifying business profiles, moderating reviews and business listings, and maintaining the integrity of the platform.

Although each module has a distinct responsibility, all modules communicate through backend services and share data stored in a centralized PostgreSQL database. This modular architecture reduces coupling between system components while improving maintainability, scalability, and ease of future enhancements.

The following figure 3.1 shows the architecture of system.

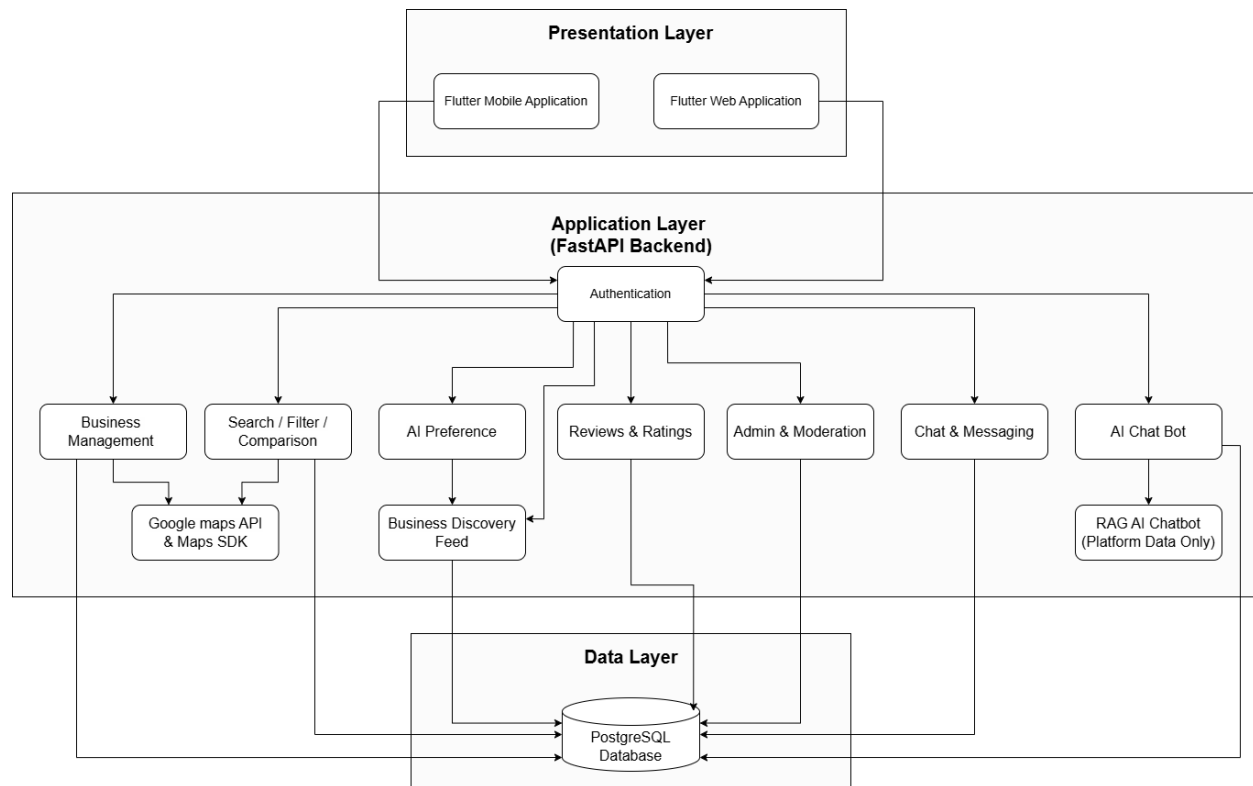


Figure 3.1 Architecture Diagram for Khojlo

The activity diagram of the Khojlo system is shown in Figure 3.2.

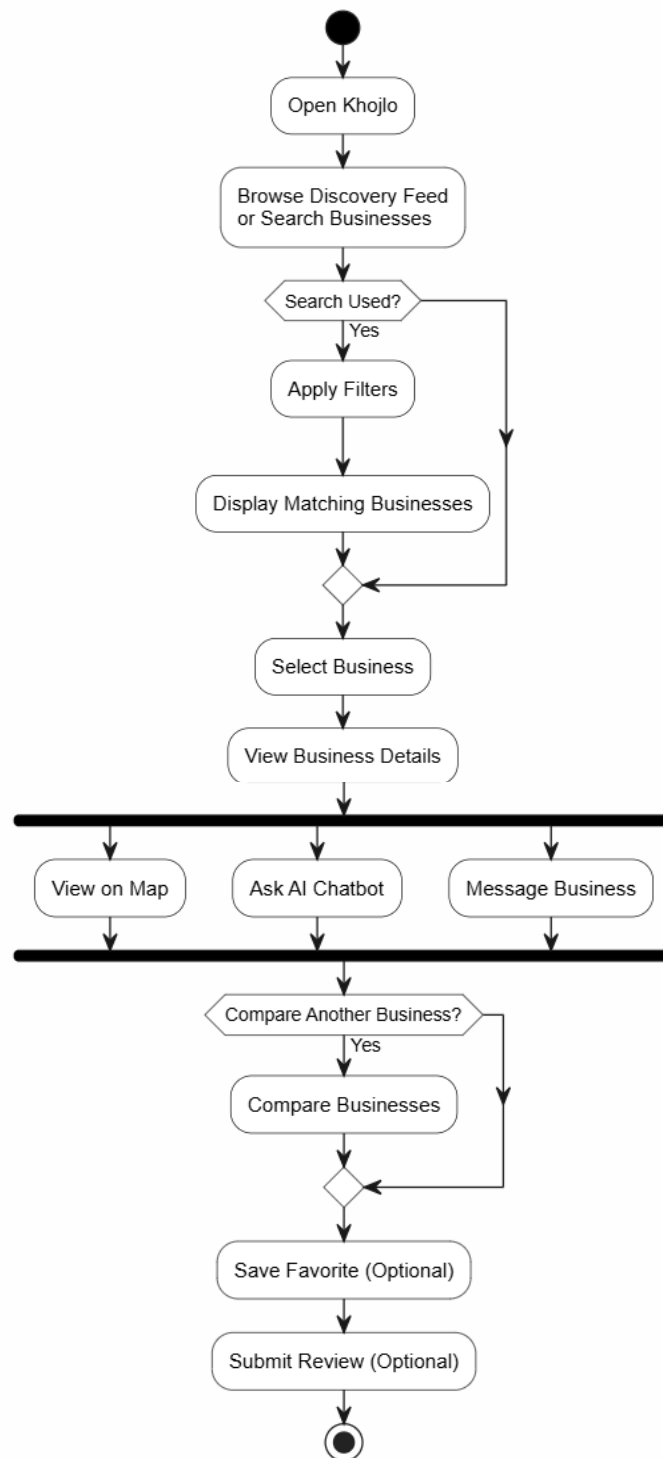


Figure 3.2 Activity Diagram for Khojlo

4.Design models

The class diagram represents the static structure of the Khojlo system by illustrating the major classes, their attributes, operations, and the relationships between them.

4.1 Class Diagram

The class diagram models the core entities of the application and demonstrates how object-oriented principles such as inheritance, association, composition, and interfaces are applied within the system.

Figure 4.1 presents the class diagram of the Khojlo system.

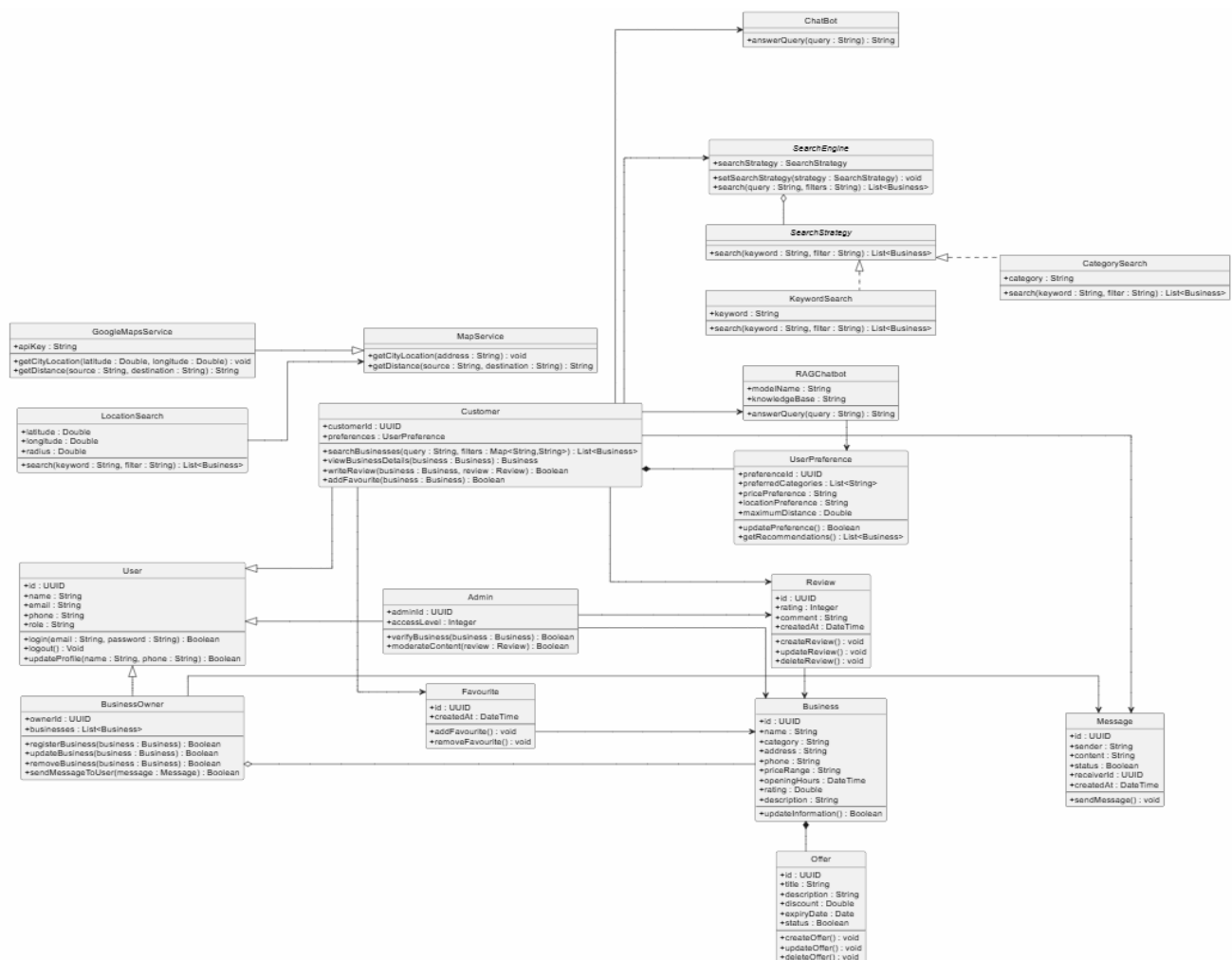


Figure 4.1 Class diagram for Khojlo

The class diagram is centered around the abstract User class, from which Customer, BusinessOwner, and Admin inherit common user attributes and operations. The BusinessOwner manages one or more Business entities, while each business may contain multiple Offers and Reviews. Customers can save businesses as favourites, submit reviews, and communicate directly with businesses through the Message class.

To support extensibility, the system uses the Strategy Pattern for implementing different search mechanisms through the SearchStrategy interface and its implementations. Similarly, the MapService and Chatbot interfaces abstract external services, allowing Google Maps and the RAG-based AI chatbot to be integrated without tightly coupling the application to a specific provider. This design improves modularity, maintainability, and future scalability.

4.2 Sequence Diagram

The sequence diagram illustrates the interaction between the customer and the major software components of the Khojlo system during the business discovery process. It demonstrates how requests are processed by the application, how data is retrieved from the database, and how external services such as Google Maps and the AI chatbot are integrated to assist the user. The diagram emphasizes the chronological order of communication between the system components while performing one of the primary functionalities of the application.

The sequence diagram of the Khojlo system is shown in Figure 4.2.

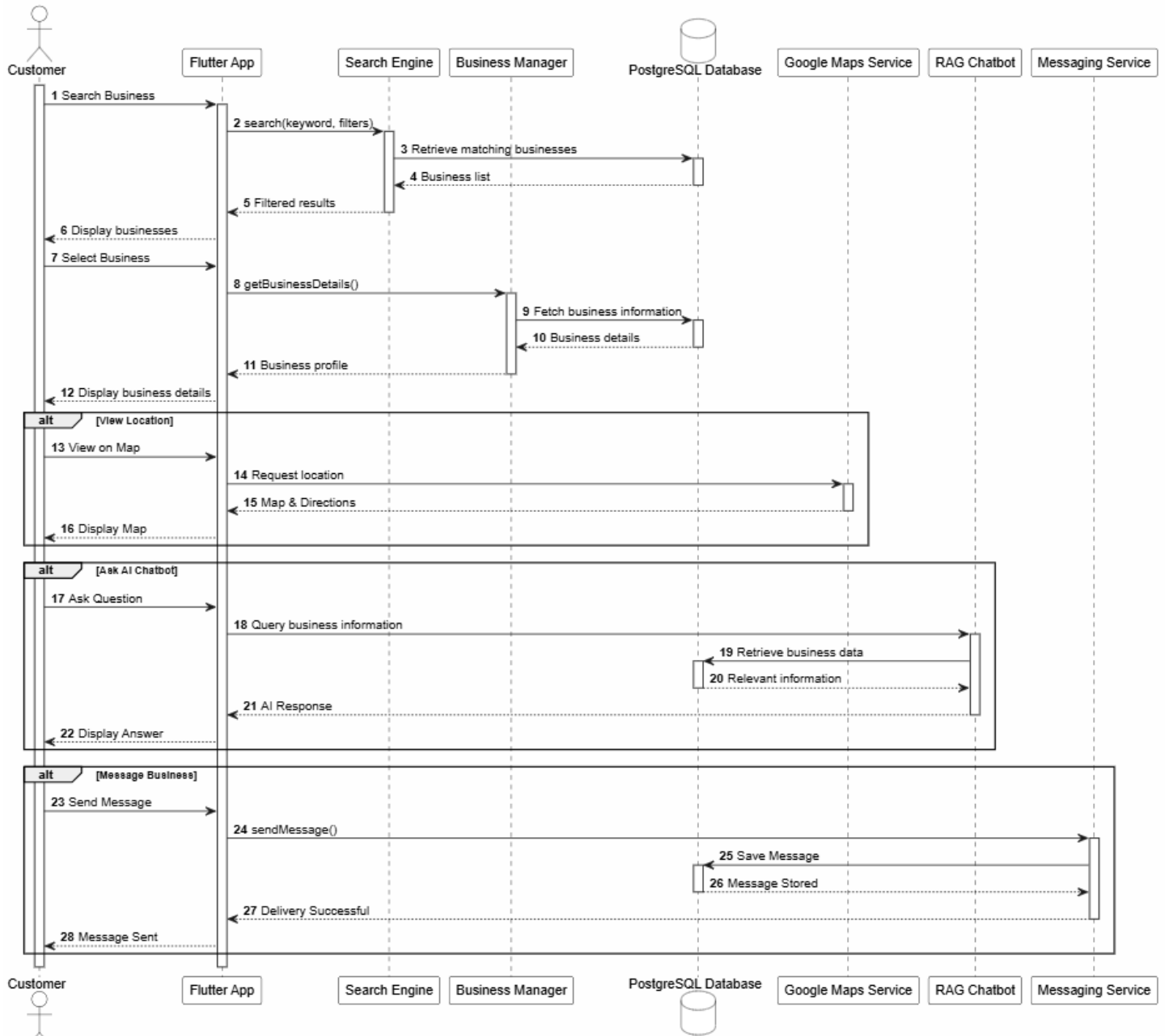


Figure 4.2 Sequence diagram for Khojlo

The sequence diagram represents the primary business discovery workflow of the Khojlo system. The process begins when the customer searches for a business through the Flutter application by providing keywords and optional filters. The search request is forwarded to the Search Engine, which retrieves the relevant business listings from the PostgreSQL database and returns the filtered results to the user interface.

After selecting a business, the application requests detailed information from the Business Manager, which retrieves the required data from the database and presents the complete business profile to the customer. From the business details page, the customer may perform additional actions depending on their requirements. The user can view the business location using the Google Maps Service, ask business-related questions through the RAG-based AI chatbot, or communicate directly with the business owner using the messaging service. Each optional interaction communicates with the appropriate subsystem before returning the requested information or confirmation to the user. This sequence demonstrates the collaboration between the application's internal components and external services to provide a comprehensive business discovery experience.

4.3 State Transition Diagram

The state transition diagram illustrates the interaction flow of a customer within the Khojlo system while discovering businesses, saving favourites, writing reviews, and communicating with business owners. It shows the different states a user passes through from being logged out to browsing businesses, viewing business details, saving businesses, submitting reviews, and initiating chat conversations. The diagram also represents the transitions triggered by user actions, providing a clear view of how customers navigate through the platform.

Figure 4.3 presents the state transition diagram

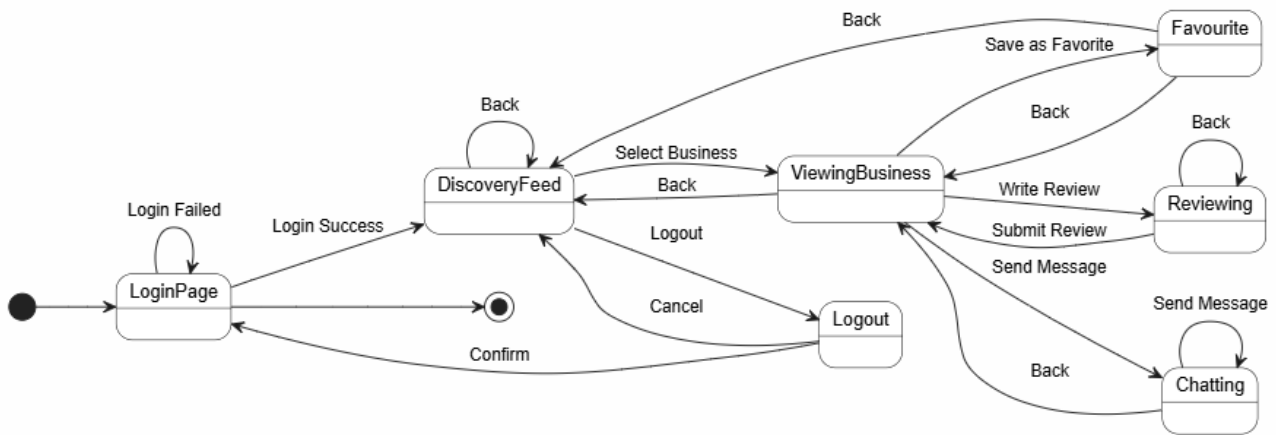


Figure 4.3 presents the state transition diagram

The state transition diagram models the interaction of a customer with the Khojlo platform. The user starts in the LoggedOut state and moves to LoggedIn after successful login. The customer can then open discovery, browse businesses, and view a selected business.

While viewing a business, the customer may save it as a favorite, write and submit a review, or send a message to the business owner. These actions lead to the Favourite, Reviewing, and Chatting states respectively. The customer can return to the business view from review or chat states using the back action. Finally, logging out transitions the user back to the LoggedOut state.

5. Data design

The Khojlo system follows a relational data model, where application data is stored and managed using a PostgreSQL database. The information domain identified during the analysis phase is transformed into normalized relational tables representing the major entities of the system. Each entity corresponds to a real-world object such as users, businesses, reviews, offers, favourites, and messages, with relationships established through primary and foreign keys to ensure data consistency and integrity.

The system stores user information in a centralized User entity, from which different user roles (Customer, Business Owner, and Administrator) are managed through role-based authorization. Business owners can register and manage one or more business profiles, while each business maintains its own details, geographical location, operational information, promotional offers, customer reviews, and ratings.

Customers interact with the system by searching for businesses, saving favourite businesses, submitting reviews, communicating with business owners through direct messages, and requesting information from the AI chatbot. These interactions generate transactional data that is stored separately to improve organization and scalability.

The application follows a client-server architecture where the Flutter frontend communicates with the FastAPI backend through RESTful APIs. The backend processes user requests, validates data, performs business logic, and interacts with the PostgreSQL database for data storage and retrieval. External services such as Google Maps are accessed through APIs and therefore do not maintain persistent application data. Similarly, the RAG-based AI chatbot retrieves relevant business information from the existing database instead of maintaining a separate application database.

The database design minimizes redundancy through normalization while maintaining efficient relationships among entities. This structure supports scalability, maintainability, and future enhancements without requiring significant changes to the existing database schema.

5.1 Database / Data Storage

The Khojlo system uses the following data storage components:

Database / Data Store	Purpose
PostgreSQL Database	Stores all persistent application data including users, businesses, reviews, offers, favourites, and messages.
Google Maps API	Provides location, map visualization, and navigation services. No application data is permanently stored.
RAG Knowledge Source	Retrieves business-related information from the platform database to generate AI responses. It does not maintain an independent transactional database for the application.

5.2 Data dictionary

Admin

Attribute	Type	Description
id	UUID	Unique administrator identifier
name	String	Administrator name
email	String	Administrator email
password	String	Encrypted password
phone	String	Contact number
role	UserRole	User role

Methods

Method	Parameters	Description
verifyBusiness()	businessId	Verifies a business profile
moderateContent()	reviewId	Removes inappropriate content

Business

Attribute	Type	Description
id	UUID	Business identifier

name	String	Business name
category	String	Business category
address	String	Business address
description	String	Business description
phone	String	Contact number
priceRange	String	Price category
latitude	Double	Latitude coordinate
longitude	Double	Longitude coordinate
status	BusinessStatus	Verification status

Methods

Method	Parameters	Description
updateInformation()	businessDetails	Updates business information

BusinessOwner

Methods

Method	Parameters	Description
registerBusiness()	businessDetails	Registers a new business
updateBusiness()	businessId, businessDetails	Updates business information
createOffer()	offerDetails	Creates promotional offers
replyMessage()	messageId	Replies to customer messages

Customer

Methods

Method	Parameters	Description
searchBusiness()	keyword, filters	Searches businesses
compareBusinesses()	businessList	Compares businesses
saveFavourite()	businessId	Saves favourite business
submitReview()	businessId, rating, comment	Adds a review
sendMessage()	businessId, message	Sends message to business

Favourite

Attribute	Type	Description
id	UUID	Favourite identifier
createdAt	DateTime	Creation date

Message

Attribute	Type	Description
id	UUID	Message identifier
content	Text	Message body
sentAt	DateTime	Sending time
isRead	Boolean	Read status

Offer

Attribute	Type	Description
id	UUID	Offer identifier
title	String	Offer title
description	String	Offer description
discount	Decimal	Discount value
startDate	Date	Offer start date
endDate	Date	Offer expiry date

Review

Attribute	Type	Description
id	UUID	Review identifier
rating	Integer	Rating value
comment	Text	Customer review
createdAt	DateTime	Review submission date

User

Attribute	Type	Description
id	UUID	Unique user identifier
name	String	Full name
email	String	Email address
password	String	Encrypted password
phone	String	Contact number
role	UserRole	User type

Methods

Method	Parameters	Description
login()	email, password	Authenticates user
logout()	—	Ends current session
updateProfile()	profileData	Updates user profile

Figure 5.1 illustrates the Entity Relationship (ER) diagram for the Khojlo

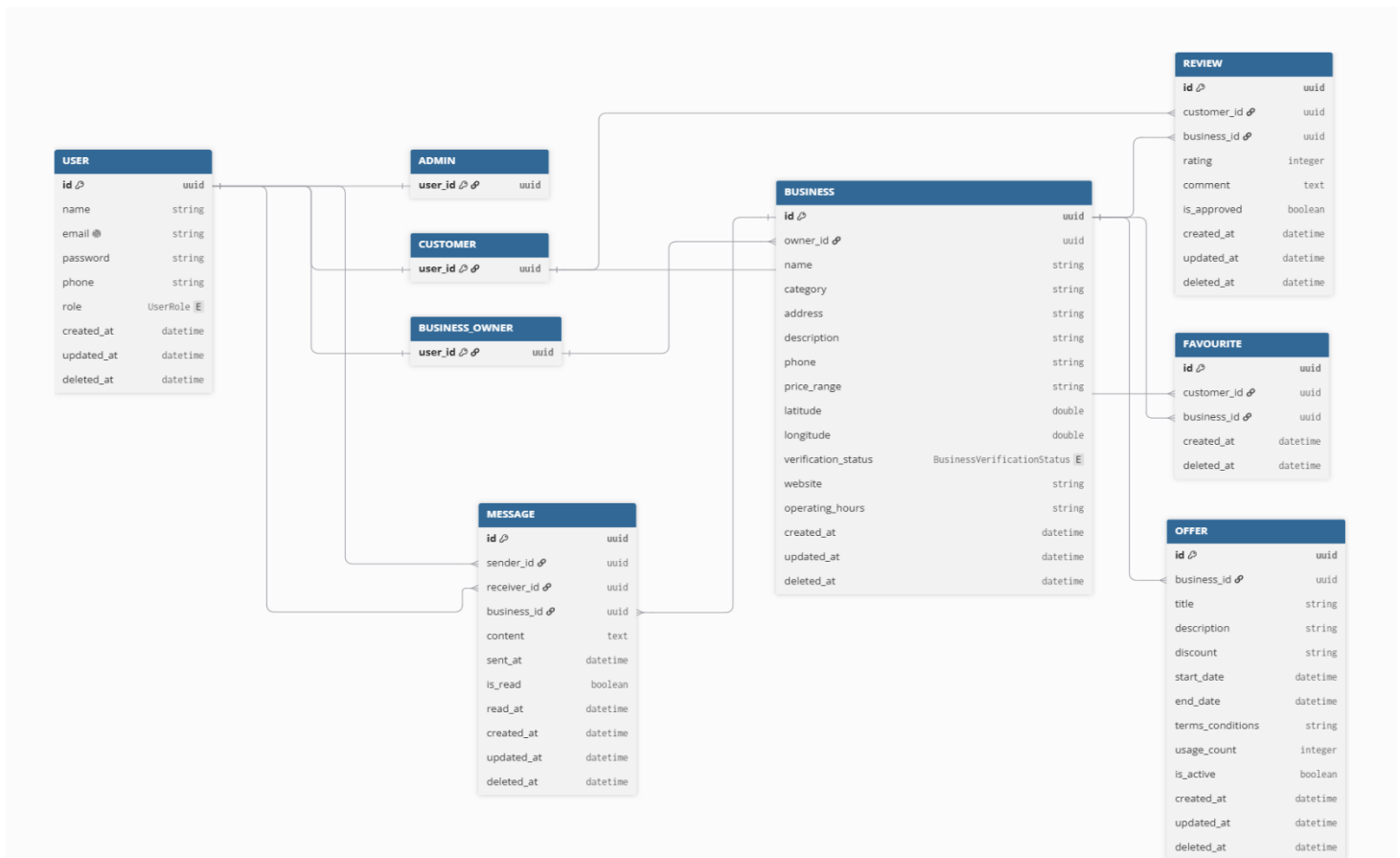


Figure 5.1 ER diagram for Khojlo

6. Algorithm & Implementation

Algorithm 1: User.login()

Description: Authenticates a registered user before granting access to the system.

Pseudo Code

BEGIN Login

INPUT email

INPUT password

IF email exists THEN

 Retrieve user record

```
    IF password matches THEN
        Create user session
        Redirect to dashboard
    ELSE
        Display "Invalid Password"
    ENDIF

ELSE

    Display "User Not Found"

ENDIF

END
```

Algorithm 2: Customer.searchBusiness()

Description: Searches businesses according to the entered keyword and selected filters.

BEGIN SearchBusiness

INPUT keyword

INPUT filters

Retrieve all verified businesses

FOR each business

IF business matches keyword
 AND satisfies filters

Add business to result list

ENDIF

ENDFOR

Sort results

Display businesses

END

Algorithm 3: Customer.compareBusinesses()

BEGIN CompareBusinesses

```
INPUT selectedBusinesses

Retrieve details of each business

Compare

    Price
    Rating
    Services
    Offers
    Distance

Display comparison table

END
```

Algorithm 4: BusinessOwner.registerBusiness()

```
BEGIN RegisterBusiness

INPUT business information

Validate information

IF validation successful THEN

    Save business

    Set status = Pending

    Notify Administrator

ELSE

    Display validation errors

ENDIF

END
```

Algorithm 5: BusinessOwner.updateBusiness()

```
BEGIN UpdateBusiness

Select business
```

Modify information
Validate data
Save changes
IF verification required THEN
 Set status = Pending Review
ENDIF
Display success message
END

Algorithm 6: BusinessOwner.createOffer()

BEGIN CreateOffer

INPUT offer details

Validate dates

Save offer

Associate offer with business

Display confirmation

END

Algorithm 7: Customer.submitReview()

BEGIN SubmitReview

INPUT rating

INPUT comment

Validate review

Store review
Recalculate average rating
Update business rating
END

Algorithm 8: Customer.sendMessage()

BEGIN SendMessage

INPUT message

Validate message

Store message

Notify business owner

Display success

END

Algorithm 9: Admin.verifyBusiness()

BEGIN VerifyBusiness

Retrieve pending business

Review business details

IF requirements satisfied THEN

 Status = Verified

ELSE

 Status = Rejected

ENDIF

Notify business owner

END

Algorithm 10: Admin.moderateContent()

BEGIN ModerateContent

Retrieve reported content

Review content

IF violates policy THEN

 Remove content

ELSE

 Keep content

ENDIF

END

Algorithm 11: SearchEngine.search()

BEGIN Search

Receive search strategy

Execute strategy

Retrieve matching businesses

Return results

END

Algorithm 12: GoogleMapsService.displayLocation()

BEGIN DisplayLocation

Receive coordinates

Request Google Maps API

Receive map

Display location

END

Algorithm 13: RAGChatbot.answerQuery()

BEGIN AnswerQuery

Receive user query

Retrieve relevant business information

Generate contextual response

Return response

END

7. Software requirements traceability matrix

Req. No.	Requirement	Ref. Item	Design Component	Component Item(s)
FR01	User Registration and Login	Class Diagram	User	login(), updateProfile()
FR02	User Profile Management	Class Diagram	User	updateProfile()
FR03	Business Registration	Class Diagram	BusinessOwner	registerBusiness()
FR04	Business Profile Management	Class Diagram	BusinessOwner, Business	updateBusiness(), updateInformation()
FR05	Business Discovery Feed	Sequence Diagram	Business Discovery Process	Search Business Workflow
FR06	Search Businesses	Class Diagram	Customer, SearchEngine	searchBusiness(), search()
FR07	Filter Businesses	Class Diagram	SearchEngine, SearchStrategy	search(), apply search strategy
FR08	Compare Businesses	Class Diagram	Customer	compareBusinesses()
FR09	View Business Details	Sequence Diagram	Business Discovery Process	View Business Details

FR10	View Business Location	Class Diagram	GoogleMapsService	displayLocation()
FR11	Submit Reviews and Ratings	Class Diagram	Customer, Review	submitReview()
FR12	Save Favourite Businesses	Class Diagram	Customer, Favourite	saveFavourite()
FR13	Business Offers and Promotions	Class Diagram	BusinessOwner, Offer	createOffer()
FR14	AI Chatbot Assistance	Class Diagram	RAGChatbot	answerQuery()
FR15	Direct Messaging	Class Diagram	Customer, Message	sendMessage()
FR16	Business Reply to Messages	Class Diagram	BusinessOwner, Message	replyMessage()
FR17	Business Verification	State Transition Diagram	Business Lifecycle	Pending → Verified
FR18	Review Moderation	Class Diagram	Admin	moderateContent()
FR19	Business Moderation	Class Diagram	Admin	verifyBusiness()
FR20	Business Status Management	State Transition Diagram	Business Lifecycle	Draft, Pending, Verified, Rejected, Suspended, Archived

8. Human interface design

Khojlo provides a simple, intuitive, and user-friendly interface that enables users to discover local businesses efficiently. The application is designed with a clean layout, consistent navigation, and responsive user interactions to ensure a seamless user experience for customers, business owners, and administrators.

When users launch the application, they are presented with the authentication screen where they can log in or create a new account. New users are guided through an AI-powered onboarding process, during which they select their preferences such as business categories and interests. These preferences are used to personalize the discovery feed and business recommendations.

After authentication, users are directed to the Discovery Feed, where they can browse newly added, trending, and recommended businesses. The application provides search, filtering, and comparison features that allow users to quickly find businesses matching their requirements. Selecting a

business opens a detailed profile displaying its description, contact information, location on Google Maps, ratings, reviews, offers, and available messaging options.

Customers can submit reviews, save businesses as favourites, compare multiple businesses, and communicate directly with business owners through the integrated messaging system. Additionally, users can ask questions using the AI Chatbot, which retrieves relevant information from the platform database using the Retrieval-Augmented Generation (RAG) approach.

Business owners have access to a dashboard where they can register businesses, update business information, create promotional offers, and respond to customer messages. Administrators access a dedicated management interface to verify business registrations, moderate reviews, and remove inappropriate or spam content.

Throughout the system, users receive immediate visual feedback for all important operations. Successful actions such as login, review submission, business registration, or preference updates are confirmed through success messages. Invalid inputs, authentication failures, or system errors are communicated using clear validation messages that help users resolve issues quickly.

8.1 Screen images

The following figures present the screen images of the Khojlo system, providing a visual representation of the application's user interface and major functional modules.

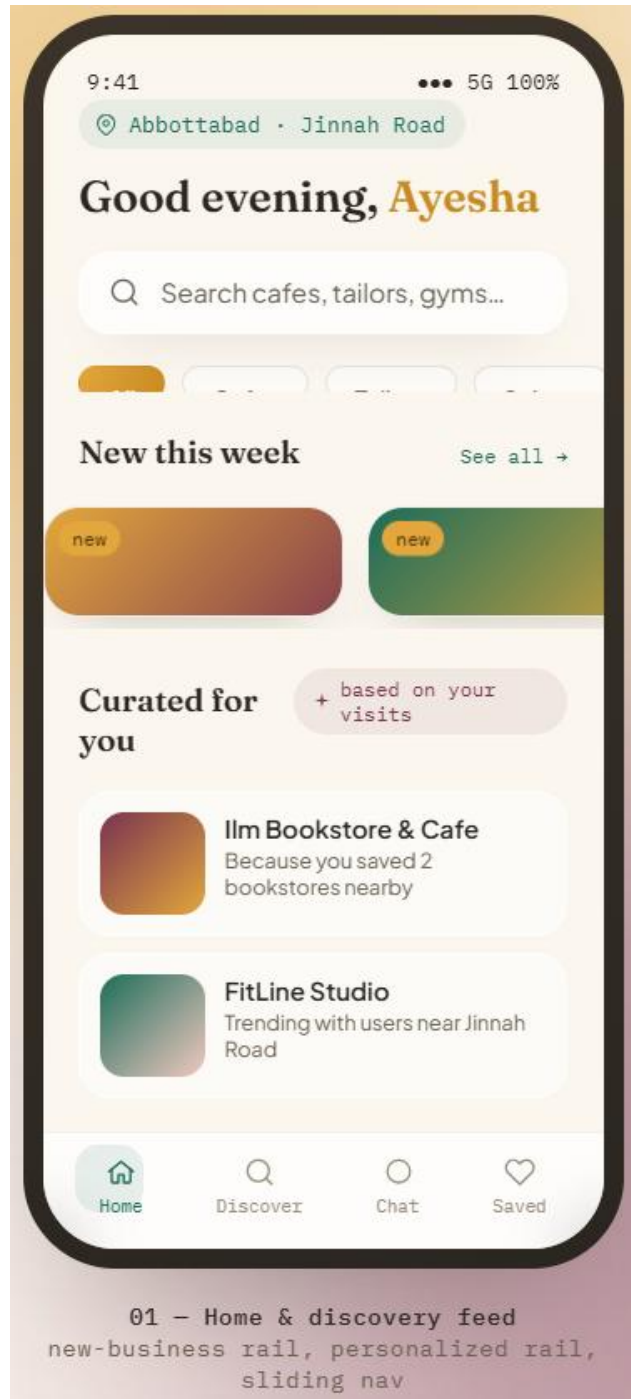


Figure 8.1

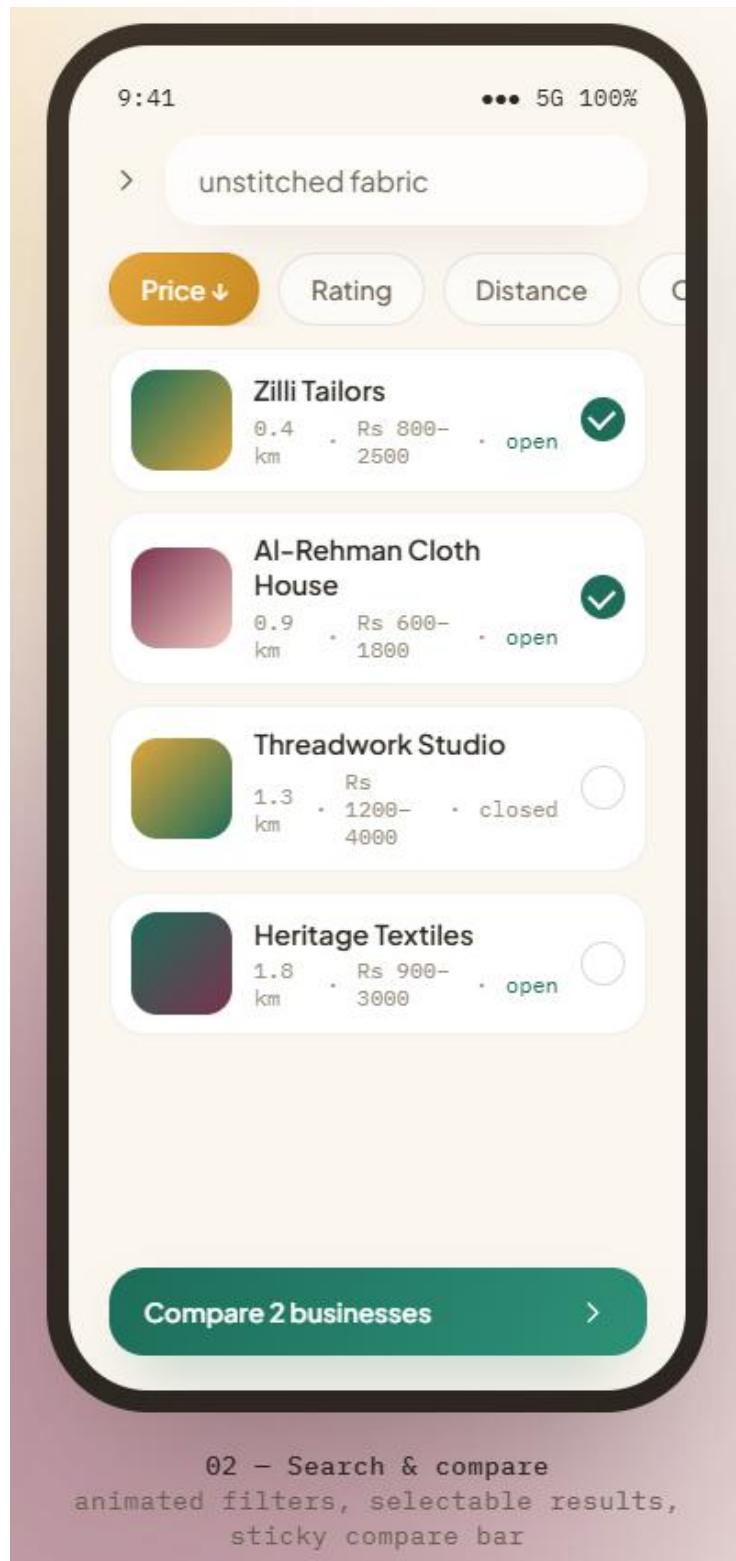


Figure 8.2

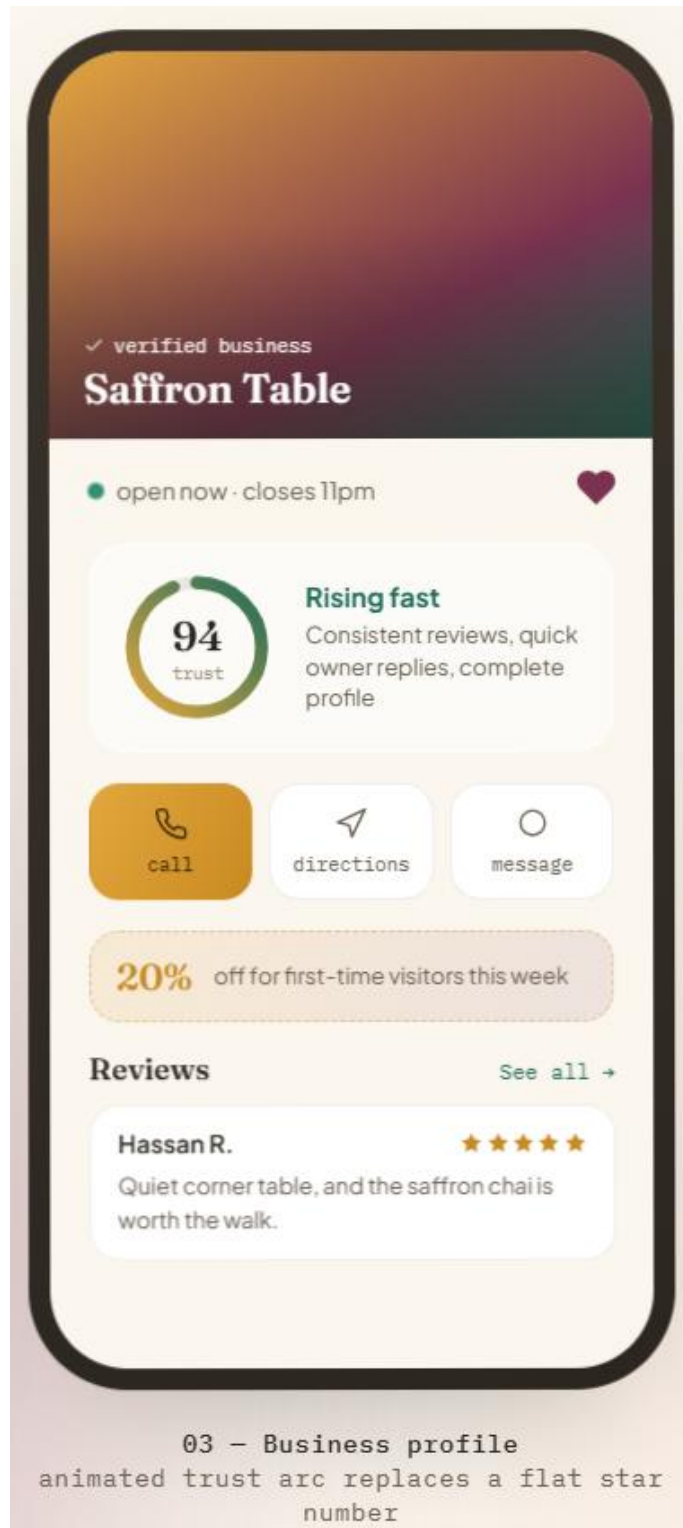


Figure 8.3

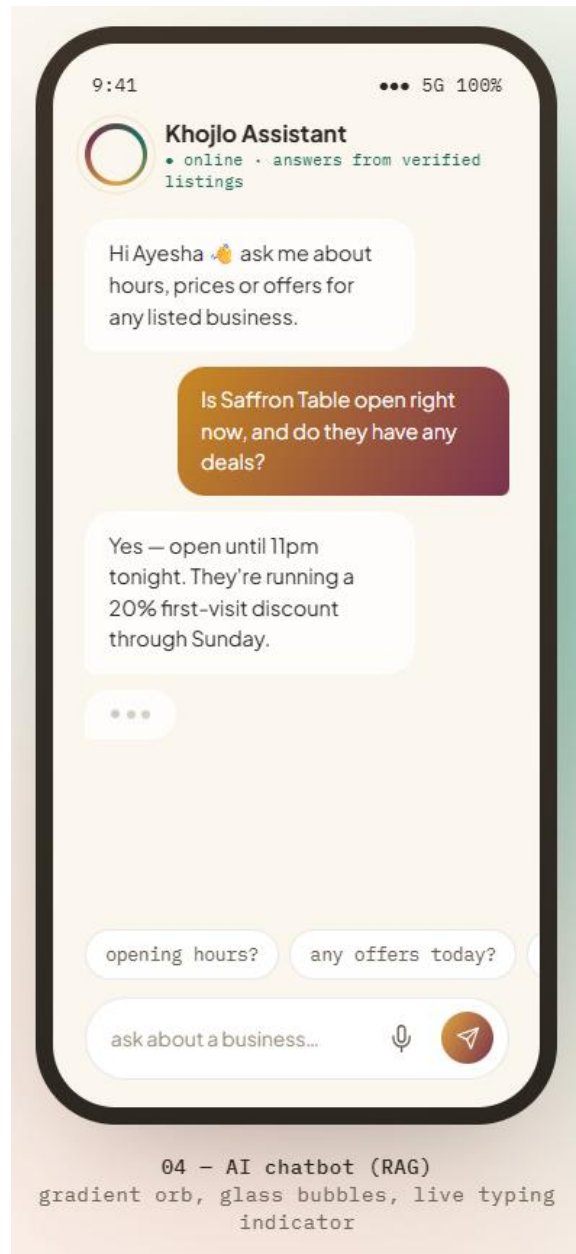


Figure 8.4

8.2 Screen objects and actions

This section describes the interface elements (screen objects) presented to the user and the actions associated with each object. The interface is designed to provide a simple, intuitive, and user-friendly experience while allowing users to efficiently discover businesses, compare services, communicate with business owners, and obtain business information through the AI assistant.

Screen 1: Home & Discovery Feed

The Home screen serves as the primary landing page after user authentication. It displays personalized recommendations, recently added businesses, and provides quick access to search and navigation features.

Screen Object	Action
Location Indicator	Displays the user's current location used for nearby business recommendations.
Greeting Section	Welcomes the user by displaying their name.
Search Bar	Allows users to enter keywords to search for businesses.
Category Chips	Filters businesses based on selected categories.
New This Week Section	Displays recently added business listings. Selecting a card opens the Business Profile screen.
Curated for You Section	Displays AI-personalized business recommendations based on user preferences and activity.
Business Cards	Displays summary information about each business. Selecting a card opens the Business Profile screen.
Bottom Navigation Bar	Provides navigation to Home, Discovery, Chat, and Saved screens.

Screen 2: Search & Compare

The Search screen enables users to search businesses, apply filters, and compare multiple businesses before making a decision.

Screen Object	Action
Search Field	Accepts keywords entered by the user to search businesses.
Filter Buttons	Filters search results based on Price, Rating, Distance, and Category.
Business Result Card	Displays business information and opens the Business Profile when selected.
Selection Indicator	Allows users to select businesses for comparison.
Compare Button	Opens the comparison view for selected businesses.

Screen 3: Business Profile

The Business Profile screen presents detailed information about a selected business and enables users to interact with the business.

Screen Object	Action
Business Banner	Displays business name and verification status.
Favorite Button	Adds or removes the business from favorites.
Availability Status	Shows whether the business is open or closed.
Trust Score	Displays the AI-generated trust indicator.
Call Button	Initiates a phone call.
Directions Button	Opens Google Maps navigation.
Message Button	Opens direct messaging with the business.
Offer Banner	Displays active promotional offers.
Reviews Section	Displays customer reviews and ratings.
See All Button	Opens the complete list of reviews.

Screen 4: AI Chatbot Assistant

The AI Chatbot screen enables users to ask questions regarding businesses listed on the platform using Retrieval-Augmented Generation (RAG).

Screen Object	Action
Chat Header	Displays chatbot identity and status.
Chat Messages	Displays the conversation.
Suggested Questions	Provides predefined questions.
Text Input Field	Allows the user to type a query.
Voice Input Button	Enables voice queries.
Send Button	Submits the query.
AI Response Area	Displays responses using verified Khojlo business data.